

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO5: Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.(BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour

QUESTIONS: 5

Total Marks:50

Annexure A

Design Task

Code of Conduct:

I K. Reshma (192565062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Reshma

Deepfake Detection and Face Authenticity Verification Platform

Introduction

Deepfake technology uses Artificial Intelligence (AI) to create fake videos, images, and audio that look real. While deepfakes can be used for entertainment, they can also be used for fraud, identity theft, misinformation, and cybercrime. A Deepfake Detection and Face Authenticity Verification Platform helps identify manipulated content and checks whether a person's face matches their real identity. Using PySpark and RDD operations, the platform can process a large number of uploaded videos and images quickly in a distributed environment.

1. System Architecture

The platform follows a distributed architecture where multiple computers work together to analyze uploaded content.

- **User Upload Module:** Users upload videos or images through a web or mobile application.
- **Data Storage:** Uploaded files are stored securely in distributed storage.
- **PySpark Processing Layer:** PySpark divides data into RDDs and processes it in parallel across worker nodes.
- **AI Detection Module:** Deep learning models analyze facial expressions, eye movement, lip synchronization, lighting, and image quality.
- **Face Verification Module:** The detected face is compared with a genuine reference image.
- **Risk Assessment Module:** Each file receives a risk score based on detection results.
- **Review Dashboard:** Suspicious files are sent to human reviewers for final verification.

This architecture improves speed, scalability, and accuracy because many videos and images are processed simultaneously.

2. Data Fusion Techniques

Data fusion means combining different types of information to improve the detection of deepfakes.

- **Visual Artifact Analysis:** Detects unnatural facial movements, blinking patterns, skin texture, and lighting differences.
- **Metadata Analysis:** Checks file creation date, editing software, device information, and location details.
- **Audio Analysis:** Compares lip movement with speech and identifies voice manipulation.
- **Face Embedding Comparison:** Extracts facial features and compares them with genuine identity data.
- **Temporal Analysis:** Examines changes between video frames to detect inconsistencies.

Combining visual, audio, metadata, and facial information increases the accuracy of deepfake detection.

3. Intelligent Agent Coordination

The platform uses intelligent agents, where each agent performs a specific task and communicates with the others.

- Detection Agent: Finds manipulated images and videos.
- Face Verification Agent: Matches the detected face with the original identity.
- Metadata Agent: Analyzes file information and editing history.
- Risk Agent: Calculates a confidence score and decides the level of suspicion.
- Review Agent: Sends high-risk files to human reviewers and stores feedback.

Working together, these agents reduce false results and improve the reliability of the platform.

4. Review Queue Prioritization

The platform prioritizes suspicious content using a risk score generated by AI models.

Risk Level	Action Taken
High Risk	Immediately sent for manual review.
Medium Risk	Added to the review queue for verification.
Low Risk	Stored for monitoring and future analysis.

Priority-based review helps security teams focus first on the most suspicious content.

5. Continuous Model Retraining

AI models should be updated regularly because new deepfake techniques are created frequently.

- Collect newly verified real and fake videos.
- Use reviewer feedback as labeled training data.
- Retrain deepfake detection models using PySpark distributed processing.
- Test the updated model for accuracy.
- Deploy the improved model to the platform.

Continuous retraining helps the platform detect new types of deepfakes and improves overall performance over time.

6. Advantages of Using PySpark and RDDs

- Processes large datasets faster using parallel computing.
- Supports distributed storage and computing.
- Handles videos, images, audio, and metadata efficiently.
- Scalable for millions of uploaded files.
- Improves performance compared to processing files on a single computer.

7. Applications

- Social media platforms to detect fake videos.
- Banking and digital payments for face authentication.
- Online examinations to verify student identity.
- Government identity verification systems.
- Cybersecurity and digital forensics investigations.

Conclusion

The Deepfake Detection and Face Authenticity Verification Platform is an effective AI-based solution for identifying manipulated media and verifying genuine identities. PySpark and RDD operations enable distributed processing of large amounts of uploaded content. By combining visual artifacts, metadata, audio analysis, and intelligent agents, the platform can accurately detect suspicious content, prioritize reviews, and continuously improve through model retraining. This makes the system reliable, scalable, and useful for real-world security applications.